



Reinforcement learning for precision grasping and safety-critical coordination in a robotic arm

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Abstract

This paper introduces a safety-focused, learning-based approach for object manipulation tasks in dynamic environments where human–robot collaboration is essential, ensuring adaptive and safety-aware operations. Our methodology employs simulation data to train and test policies using two different simulation frameworks: Handoversim, a recently developed benchmark for human-to-robot object handovers, and our custom simulation framework. Both frameworks are integrated with the OpenAI Safety Gym library, which supports three mobile agent types learning goal-directed tasks under various safety constraints but did not initially include a robotic arm. We have integrated a new model, the Panda robotic arm, and validated its applicability within Safety Gym. Our focus is on accurate and secure object manipulation, particularly in human–robot interaction scenarios, within the custom simulation framework end-to-end policy achieved successfully. Handoversim, while advanced, struggled with object handling due to inherent challenges. This led to established baselines switching to hand-coded policies for grasping instead of relying on end-to-end policy learning. We compared a model-free proximal policy optimization (PPO) baseline with its constrained variant (cPPO) within a reinforcement learning framework. To accelerate learning while maintaining safety compliance, we introduce "Grasp Mechanics", an innovative tactile feedback framework that reduces dependence on complex vision-based systems. This approach emulates human-like object interaction through proprioceptive feedback, enabling precise and secure manipulation without requiring detailed prior object knowledge. Preliminary experiments indicate that although cPPO requires longer training periods, it achieves policy performance comparable to the baseline PPO, while significantly outperforming it in terms of safety compliance. This paper highlights the potential of integrating advanced reinforcement learning techniques with robust safety mechanisms, as provided by Safety Gym, to elevate the capabilities of robotic systems in safely handling safety-critical manipulation tasks, crucial for real-world applications. Our findings lay the groundwork for future advancements in safe autonomous robotics, emphasizing the critical role of integrating safety considerations from the ground up.

Keywords Safe reinforcement learning · Robotic grasping · Autonomous learning · Safety-critical coordination

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1 Introduction

Significant breakthroughs in human–robot interaction have been achieved in various routine collaborative activities [1, 2], including handing over objects [3], coordinating safety-critical tasks, setting up tables [4], and preparing meals. To automate these tasks effectively, a safety-instrumented environment and intricate coordination are essential for successful human–robot collaboration. This is challenging because the robot must react to human behavior while relying on sparse raw sensory inputs, such as a single camera with a limited field of view, often compounded by issues like occlusion and latency. Exploring safety within deep reinforcement learning (DRL) has become crucial for advancing autonomous systems across various sectors. The promise of DRL lies in its ability to adapt and learn from interactions, enhancing the capabilities and resilience of autonomous systems. The fundamental principle of RL is that an agent learns to maximize a reward signal through trial and error. RL is particularly suited for problems where evaluating behaviors via a reward function is more feasible than generating optimal behaviors through analytical or numerical methods. This adaptability makes RL attractive for applications such as self-driving cars [5], surgical robotics [6–8], and energy systems management [9, 10]. Most RL successes have been achieved using simulators, allowing agents to experiment without real-world consequences. However, simulators may not always be available or sufficiently accurate for RL to learn behaviors that transfer well to real environments. While “sim-to-real” transfer learning algorithms can mitigate this issue, we anticipate that in scenarios involving AI–human interaction or highly complex systems, the challenge of building useful simulators will necessitate direct real-world training. This approach, however, poses risks of negative impacts on the environment. Motivated from the aforementioned challenges, this article demonstrates the importance of platforms like Safety Gym [11] to facilitate safe exploration in DRL through constraints. This platform ensures learning happens within safe boundaries, essential for applications like autonomous vehicles and robotics. The challenge of safe exploration is heightened by the dynamic nature of environments where robots operate, particularly in complex human–robot interactions. To address the challenges of ensuring safety in RL, OpenAI developed the Safety Gym library a comprehensive suite of environments and tools designed to facilitate the development, benchmarking, and evaluation of RL agents that adhere to safety constraints during both training and testing phases. Safety Gym offers three distinct agent types (point, car, and quadruped), various tasks (goal, button, push), and a range of safety constraints (hazards, vases, etc.). These tools enable the creation of diverse, safety-focused environments. We integrated two simulation

frameworks within Safety Gym: Handoversim framework [3] and our custom simulation.

- (1) In our custom simulation environment, we integrated a new model, the Panda robotic arm, into the OpenAI Safety Gym framework. Despite significant challenges during the integration process, we ensured its applicability, allowing us to develop a novel safety-aware environment specifically designed for training and testing constrained RL algorithms for safety-critical realistic object manipulation tasks, supported by reward and cost functions tailored to the specific requirements and contexts of use. To explore autonomous safety-critical scenarios in unpredictable settings, we emphasize the integration of robot maneuver planning around obstacle with grasping strategies to achieve more effective and safe object manipulation. Our approach focuses on stringent safety measures, balancing end-to-end learning policies while minimizing the risk of improper actions in dynamic environments. Despite these advancements, challenges remain, particularly in 6D grasping [12], which requires precise control of the robot’s gripper in space.
- (2) The Handoversim simulation framework, despite being feature-rich and offering high-fidelity simulation, still struggles with fully autonomous end-to-end policy learning, particularly in human-to-robot object handover scenarios. Handoversim equips object models with additional actuators at their base, allowing their 6D pose to be directly controlled by simulation controllers. This enhancement enables the direct control of the human hand and the grasped object according to their captured trajectories [13]. However, due to noise in the motion capture data, moderate interpenetration between the hand and object models occurs during grasping. To avoid artifacts caused by unstable simulations, Handoversim disables collision detection between the human hand and the object. Despite these efforts in providing a high-fidelity simulator, it often still requires switching to hand-coded policies [14] for stable grasps, rather than achieving fully autonomous policies as desired. Learning to control a dexterous hand to manipulate objects has been notoriously challenging [15–17] and requires significant engineering effort on the hand’s physics model. Consequently, human-to-robot object handovers necessitate direct learning with a focus on stringent safety controls to mitigate potential real-world risks. While prior environments such as Panda-Gym [18] enable goal-conditioned learning on the Panda robot, they primarily focus on short-horizon tasks and lack explicit consideration of safety—an essential requirement for real-world human–robot collaboration. In contrast, we introduce

a novel closed-loop grasping and long-horizon motion planning policy, composed of hierarchical sub-policies, which leverages both proprioceptive and tactile feedback. This design allows the robot to adapt dynamically to real-time object interactions without relying on high-dimensional sensory inputs such as images.

Our central contribution lies in the use of a hierarchical reinforcement learning (HRL) framework [1], which decomposes complex grasping behaviors into structured subgoals. This not only reduces the overall learning complexity but also promotes generalization and robust performance across diverse objects and task conditions. Furthermore, integration with the Safety Gym environment enables principled evaluation of safety-aware behavior, advancing the field toward reliable and efficient human–robot interaction in practical, real-world settings.

Our research revealed a surprising finding: robots can manipulate objects of varied geometries in dynamic environments without image-based sensory data. Initially, we anticipated the challenge would be crafting a visual processing architecture. However, our experiments demonstrated the effectiveness of control strategies like 'Safety Gym demonstration on a robotic arm' and 'Grasp Mechanics for adaptive gripping'; our system enabled both safe and efficient object manipulation. This suggests visual data may not be as crucial as assumed, though integrating high-dimensional raw information could enhance performance. This prompts a paradigm shift toward a vision-based system, as discussed in teacher–student training framework [19]. In this framework, a teacher policy is first trained using privileged information and then guides a student policy that relies solely on real-world sensory inputs. Significant progress in robotic manipulation can be achieved without sophisticated 3D vision systems, as discussed in the teacher–student training framework. This suggests that, while valuable, vision serves more as an enhancement the "icing on the cake" rather than being fundamental to the robot's operational success.

Video Demonstration of Rigorous Simulation-Based Experiments: <https://youtu.be/8K3nxhRS0Mk>

1.1 Related works

Our research builds on the foundation of Safety Gym to enhance robotics applications in unstructured environments, such as retail settings and public spaces. Recognizing the challenges of ensuring safe interactions, we draw upon the consensus that safe behavior in robots equates to avoiding hazardous states, a perspective supported by [20, 21] and further elaborated through various safety and learning strategies in the literature. Significant contributions to safety in RL include the development of constraints to prevent unsafe

actions, as suggested by [22, 23], and the incorporation of concepts like reachability and stability in algorithm design. Advanced techniques emphasize aligning robot actions with human preferences for enhanced safety [11, 24], explored through ergodicity criteria and the mitigation of adverse "side effects" in agent behaviors.

In addressing the intricacies of robotic grasping and task achievement, our study is inspired by recent advancements that combine learning from demonstration (LfD) and reinforcement learning to enable robust and adaptable policies. While LfD has shown promise—particularly in seeding policy learning with expert priors for complex manipulation tasks [25], it also presents challenges, notably the high cost and effort required for real-world data collection [26]. Future work may explore strategies to mitigate this cost, such as leveraging simulated demonstrations or hybrid frameworks that combine sparse demonstrations with reinforcement learning. This hybrid approach, along with advancements in deep learning, offers new pathways for robotic control and manipulation, particularly in the realm of object grasping where recognizing and handling objects of various geometries remains a complex challenge.

Recent works have explored deep reinforcement learning techniques for manipulation tasks using both model-based and model-free paradigms. Guided policy search, for instance, has enabled the training of visuomotor policies directly from raw images [27], although such approaches often require substantial human effort for data collection [26]. Model-based methods that learn global or local dynamics from data [28] struggle with tasks involving discontinuities or sparse rewards [29]. Other lines of research pursue large-scale, self-supervised learning [30] or simulation-based training [31, 32], but high-fidelity sim-to-real transfer often demands visual domain adaptation [33]. In contrast, many industrial and structured environments can be effectively addressed using low-dimensional kinematic state features, which are more robust to simulation discrepancies and sufficient for tasks where part locations are known in advance.

While several open-source environments, such as PandaGym [18], support goal-conditioned tasks with the Panda robotic arm, they primarily focus on performance metrics in short-horizon, simplified tasks and do not explicitly incorporate safety considerations. These platforms typically assume controlled environments and overlook challenges associated with safe exploration or constraint satisfaction. In contrast, our work builds on the Safety Gym framework to embed safety directly into the learning process. It further extends the scope to long-horizon, multi-object interactions enhanced by Grasp Mechanics—leveraging tactile feedback, and safety-critical tasks such as setting up tables or performing assistive functions, which are essential for real-world deployment in dynamic and uncertain environments.

Recent innovations in DRL, including policy search methods like PPO and its constrained variant, highlight the evolving capabilities of robotic systems in learning and executing tasks with high efficiency and rigorous safety control. Our research aims to harness these advancements to develop strategies that enable robots to grasp and manipulate objects safely in complex and unstructured environments, pushing the boundaries of what's achievable in robotic autonomy.

1.2 Our contributions are as follows:

- **Safety Gym Demonstration:** We introduce a new robot arm (agent), to the Safety Gym framework, showcasing a novel layout for realistic object manipulation scenarios with a focus on safety-critical operations.
- **cPPO Advantages:** Through a rigorous comparison with standard PPO, demonstrated cPPO's capability to maintain safety constraints during robotic manipulation, while achieving comparable task performance.
- **Adaptive Grasping Policy:** We developed a closed-loop, goal-directed grasping policy that leverages proprioceptive and tactile feedback for real-time adaptation to varied objects, contrasting with previous approaches that rely on predefined policies or offline planning methods [14, 34]
- **Zero-Shot Transfer Policy:** We evaluate the zero-shot transfer policy on unseen objects, showcasing the adaptability and robustness of our approach.
- **Safety Gym on Handoversim Simulation Framework and Benchmark:** Our supplementary study integrated the Safety Gym framework with the Handoversim benchmark, enhancing safety in human-to-robot object handovers. The findings highlight the potential of safe RL methods to mitigate real-world risks and improve policy learning for safety-critical tasks.
- **Prototype Development:** Our approach presents a novel prototype that reduces reliance on complex vision-based systems by leveraging tactile feedback, mirroring human-like interaction with objects in real-world environments.

Our approach addresses a critical gap in robotic manipulation by ensuring strict collision avoidance while preserving task performance, a balance often neglected in existing systems that prioritize task success at the expense of operational safety, as demonstrated in the accompanying video.

1.3 Finding Technical Solutions

- **Simulation Frameworks Integration with Safety Gym:** On the one hand, the availability of numerous Python resources and simulators built on various physics engines offers significant advantages. On the other hand, integrating them can often be challenging. Our primary

motivation was to demonstrate constrained RL algorithms using a robot arm (agent) for use in human–robot interaction scenarios, which is not included in the Safety Gym library's agents [11]. This integration posed significant challenges. Safety Gym is developed using the MuJoCo simulation engine [35], so our initial approach was to import a robotic arm from Gym environment into Safety Gym framework, expecting compatibility due to their shared MuJoCo foundation. However, we encountered numerous technical challenges, including dependency challenges and the necessity to utilize earlier versions of Python and TensorFlow. We also explored integration with CoppeliaSim via PyRep but faced similar compatibility issues. Ultimately, we adopted a PyBullet-based Gym environment [36], which supports robotic arms and provided comprehensive documentation and examples available online, which greatly facilitated setting up the environment in the desired way.

2 Problem formulation

To simulate the physical interaction of this process, Our research employs the Safety Gym framework to implement safety controls, we built a simulation environment using the PyBullet physics engine. We chose the Franka Emika Panda robot, which features a 7-DoF arm and a 2-DoF parallel-jaw gripper. The intended task involves a tabletop scene where the robotic arm maneuvers objects. This scene consists of the robotic arm, an obstacle that appears randomly in close proximity, and a target object to be secured into a cabinet. This task is modeled as a constrained Markov decision process (CMDP) [11], a framework that describes the interaction between an agent and its environment, incorporating an additional cost function to manage trade-offs between cumulative return and hazard avoidance. Constraints are essential for formulating safety requirements, aiming to avoid harm by preventing hazardous states and behaviors [37]. This approach is fundamental to systems safety standards. Unlike standard RL, which maximizes a reward function, safe agent behavior requires balancing rewards with costs for proximity to hazards. Incorrect cost selection can lead to unsafe behavior or hinder learning. This method enhances the safety and efficiency of robotic maneuvers in a simulated environment, laying the groundwork for real-world applications. Safety Gym environments are evaluated based on three metrics: task performance of the final policy, constraint satisfaction, and average regret regarding safety costs during training.

2.1 Pretraining phase

- The primary objective of pretraining phase is to encode task and safety criteria and develop an instrumented train-

ing environment that represents these requirements and their context of applicability. The key components of policy learning are outlined as follows:

- Intended Functionality is in contrast to traditional software systems, it is difficult to specify or verify an ML skill's whole behavior with precision [38]. The ISO/PAS 21448:2019 standard for road vehicles introduces the term "intended functionality" to define the requirements specification. Intended functionality encompasses the task definition and the context of applicability expected to be covered by the ML skill.
- Expert knowledge is crucial for creating an instrumented training environment that encodes both the task and the context of applicability into a RL setting. This expertise is demonstrated in two key ways: (1) by providing instrumented training environments that accurately represent the context of applicability and (2) by encoding task and safety requirements into reward functions [39] and cost functions [40, 41].
- Regarding the action representation, it is defined by a vector $[dx, dy, dz]$, representing changes in the position of the arm's tip in 3D Cartesian space, and a Boolean for the gripper's status (open or closed), totaling four features. These values are used to calculate the new position of the tip, and inverse kinematics determine the necessary joint adjustments [16]. The agent's operation relies on collecting low-dimensional proprioceptive data to inform its actions, focusing on precise movements along the X, Y, and Z axes. By leveraging inverse kinematics solver ensures accurate positioning and differential movement of the robotic arm's end-effector within a 3D space.
- The state representation includes the robot's end-effector position in 3D Cartesian space, the object's position, and the status of the fingers, totaling a vector of 8 features. These aspects are enhanced by Grasp Mechanics, enabling precise and adaptive control actions while avoiding collisions.

3 Methodology

3.1 Safe reinforcement learning through constraints

Introduction to Constrained RL: Constrained RL [11] is an advanced methodology focused on training RL agents to operate within predefined safety constraints, ensuring safe behavior both during training and deployment. This approach is particularly vital in scenarios where safety is paramount. An optimal policy in safe RL is defined as:

$$\pi^* = \arg \max_{\pi \in \Pi_C} J_r(\pi) \quad \text{where} \quad \Pi_C = \{\pi : J_{c_i}(\pi) \leq d_i, i = 1, \dots, k\} \quad (1)$$

where $J_r(\pi)$ represents an objective function driven by rewards and each J_{c_i} signifies a cost-oriented constraint function, with thresholds d_i chosen as hyper-parameters by humans. This constraint mechanism defines a valid set of permissible policies Π_C that meet the constraints, within the framework of constrained Markov decision processes (CMDPs). The Lagrangian technique employs a composite loss function, integrating both reward-based and cost-based elements. In the equation below, cost-related component is incorporated within the space of allowable policies Π_C . The optimization problem is formulated as follows:

$$\max_{\theta} \min_{\lambda \geq 0} L(\phi, \theta) = f(\theta) - \lambda g(\theta) \quad (2)$$

Within the loss function, the two components correspond to the reward and the cost, incorporating the policy model parameters θ and the Lagrangian hyperparameter λ . Building on the pioneering work of [19] in the teacher and student training framework and others who demonstrated the potential of model-free RL in complex hand-object interactions, our study advances the use of low-dimensional privileged information for policy training with safety constraints in robotic tasks. By focusing on direct learning from enriched simulated data, our approach aims to refine task manipulation strategies for varied object geometries without the need for intricate modeling.

3.2 Task design and objectives

To evaluate the capabilities of intelligent systems in structured manipulation scenarios, we introduce two hierarchically related tasks with escalating complexity. The first task, termed single-object phased manipulation (SOPM), requires an agent to execute a sequence of tightly coupled subgoals to transport a single object to a target location. This task is characterized by its long-horizon structure, where the agent must sequentially coordinate approaching, grasping, lifting, positioning, and releasing the object within a unified workflow. Unlike atomic primitives such as bin pick-and-place, SOPM emphasizes temporal coherence across distinct phases, necessitating robust error recovery and intermediate state tracking to avoid cascading failures. Building upon this foundation, the second task, multi-object atomic manipulation (MOAM), introduces combinatorial complexity by requiring an agent to plan and execute disposition trajectories for three objects in a single episode. Here, success depends on resolving interdependent spatial reasoning challenges, such as maneuvered actions at each trajectory point, thus leading to a successful episode. MOAM maintains all the adaptive and dynamic feedback mechanisms of SOPM while adding the atomic constraint that all objects must be positioned correctly, thereby testing an agent's ability to bal-

ance global efficiency with local precision. This task extends SOPM's phased paradigm into a multi-object domain, where the interplay between object trajectories introduces emergent challenges absent in single-object scenarios. Together, these tasks provide a framework for evaluating how systems manage both intra-object phase transitions (SOPM) and inter-object coordination (MOAM), without episodic resets, thereby testing real-time decision-making, hierarchical planning, and robustness in real-world manipulation.

3.3 Proposed approach structure

Our methodology introduces a constrained DRL strategy, centering on proximal policy optimization (PPO) and its constrained variant (cPPO). These techniques are evaluated using the Safety Gym benchmark to ensure their viability in simulated scenarios and to highlight their potential for real-world applications. This benchmarking is crucial for assessing the effectiveness of our strategy and its adherence to safety standards (Fig. 1).

3.4 Key components

- **State Space** Our framework seamlessly integrates simulation data with proprioceptive state information, encompassing the following attributes within Cartesian space: robot states (forces exerted on the robot end-effector), fingertip states, and object states. Additionally, it includes data from collision sensors, visual sensors, and distance measurements. These elements collectively provide tactile feedback for adaptive gripping. This comprehensive set of state information ensures a continuous state space, which is crucial for the effective learning of manipulation tasks. In continuous state spaces, deep neural networks (DNNs) excel in learning low-dimensional representations that efficiently map states to optimal control actions, thereby enhancing the robustness and precision of robotic manipulation, as facilitated by the Grasp Mechanics.
- **Action Space** In human–robot collaboration (HRC) scenarios, the action space is defined by complex interactions between humans and robots in a six-dimensional space (the robot's end-effector pose). This continuous action space presents a significant challenge for RL agents, requiring extensive training times, often spanning millions of time steps, even when utilizing robust exploration and exploitation algorithms [42]. To address these challenges, recent approaches have focused on constraining the action space by leveraging existing inverse kinematics (IK) solvers rather than directly learning the joint angles of the robot [23]. This method effectively reduces the action space dimensionality from seven dimensions (corresponding to the seven degrees of freedom (DoF) of robot arms) to three dimensions,

considering only positional changes in Cartesian space (Δx , Δy , and Δz) with fixed orientations. This reduction not only expedites the learning process but also enhances task efficiency and safety by guiding continuous learning. Additionally, learning on preexisting skills results in more robust output behaviors, minimizing the likelihood of unintended actions. Moreover, this approach facilitates the transferability of learned behaviors across different types of robots, thereby promoting generalization and adaptability in diverse HRC applications.

- **Reward Function** The reward function acts as a virtual coach, guiding the robotic arm in object manipulation tasks toward the target goal, ensuring precise movement and obstacle avoidance. Negative reward functions act as soft penalties, which may not guarantee safety. Thus, constrained RL maximizes expected returns while adhering to safety constraints using cost functions [11], ensuring safe goal achievement by limiting exploration and optimizing actions within the constrained space. Designing an effective reward function is complex and requires significant effort, constituting approximately 60% of the total development process. This effort is crucial to encourage desired behaviors, mitigate unintended actions, and directly influence the robot's learning efficiency and task performance.

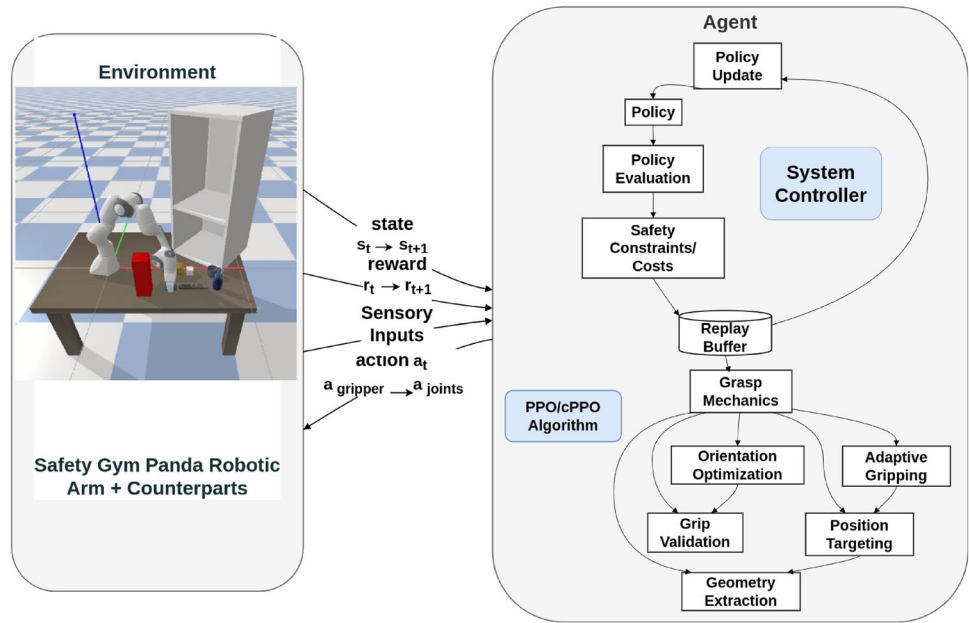
3.5 Hierarchical task execution via high-level phase control

We adopt an HRL framework to decompose complex multi-object manipulation tasks into a sequence of semantically meaningful phases or subgoals. Each phase is associated with its own reward function, which guides the learning process in a localized and context-sensitive manner. This hierarchical decomposition improves both learning efficiency and execution robustness by enabling the agent to focus on solving simpler, temporally extended sub-tasks in a sequential manner, rather than tackling the full task end-to-end. Inspired by the options framework [1], our high-level policy acts as a controller over temporally extended sub-policies, or macro-actions, each governing a specific phase of the task: approach, place, retract, and wait. This modular decomposition enables interpretability, sample-efficient learning, and scalability to varied manipulation scenarios.

Formally, the high-level policy is defined as a mapping from the current action and state to a phase-specific low-level policy:

$$\pi_{\text{high}}(a_t, s_t) = \pi_{\phi(t)}(a_t), \quad (3)$$

where a_t denotes the agent's action at time t , s_t is the current system state, and $\phi(t) \in \{\text{approach, place, retract,}$

Fig. 1 Agent Equipped with Grasp Mechanics

wait, *done*) indicates the active phase. Each sub-policy $\pi_{\phi(t)}$ operates within its own temporal abstraction, and transitions between phases are governed by termination conditions—such as contact detection, grasp success, or target arrival—mirroring the option termination criteria in HRL.

In the *approach* phase, the robot navigates toward a target object using a control policy based on the end-effector's pose and a relative action vector. To ensure safe descent, we define a hover-aware control law that adjusts vertical motion as a function of the current height z_t and a predefined hover height z_{hover} :

$$z_{t+1} = \begin{cases} z_t - \alpha |a_z|, & \text{if } z_t > z_{\text{hover}}, \\ z_t + \beta (z_{\text{hover}} - z_t), & \text{otherwise,} \end{cases} \quad (4)$$

where α and β are tuning coefficients controlling descent and hover smoothing.

Upon a successful grasp and lift, the system transitions to the *place* phase, where the end-effector follows a linear interpolation trajectory toward a predefined target location:

$$\mathbf{x}_{t+1} = (1 - \lambda_t) \mathbf{x}_{\text{start}} + \lambda_t \mathbf{x}_{\text{target}}, \quad \lambda_t \in [0, 1], \quad (5)$$

where λ_t is an incrementally increasing interpolation factor. When the target is reached, the object is released and the system enters the *retract* phase.

In the *retract* phase, the gripper moves to a safe retraction position and transitions to the *wait* state upon reaching the home configuration:

$$\phi(t+1) = \begin{cases} \text{waiting,} & \text{if home reached,} \\ \text{retracting,} & \text{otherwise.} \end{cases} \quad (6)$$

During the *wait* phase, the system remains idle until a new target object is detected, at which point the *approach* phase is reinitialized; otherwise, the system transitions to a terminal *done* state:

$$\phi(t+1) = \begin{cases} \text{approaching,} & \text{if new object found,} \\ \text{done,} & \text{otherwise.} \end{cases} \quad (7)$$

This hierarchical structure embodies the central tenets of HRL: temporal abstraction, policy modularity, and state-dependent transitions. It allows for efficient reuse of learned behaviors across different task stages, simplifies exploration by narrowing the action space within each phase, and provides a natural foundation for policy learning via deep hierarchical RL methods such as the Options-Critic architecture [43] or FeUdal Networks [44].

4 Implementation settings

In our study, we employ the proximal policy optimization (PPO) algorithm, which improves upon the policy gradient (PG) method by enabling limited reuse of collected experience while retaining its on-policy characteristics. This approach allows multiple policy updates per data collection phase while preventing large policy changes that could destabilize training. The PPO algorithm uses importance sampling with an objective function based on the advantage function $A(s_t, a_t)$, facilitating efficient policy updates. We adopt the PPO-Clip variant, which introduces a clipping mechanism in the objective function to constrain policy updates within specified bounds, ensuring stable learning. The objective function is defined as:

$$L(s, a, \theta_k) = \min \left(\frac{\pi_{\theta}(a|s)}{\pi_{\theta_k}(a|s)} A^k(s, a), \right. \\ \left. \text{clip} \left(\frac{\pi_{\theta}(a|s)}{\pi_{\theta_k}(a|s)}, 1 - \epsilon, 1 + \epsilon \right) A^k(s, a) \right) \quad (8)$$

This approach moderates the policy updates, guiding them toward an optimal policy that maximizes the reward while controlling excessive deviation from the previous policy.

To address safety concerns within the RL framework, our work integrates the PPO algorithm with the Lagrangian method for constrained optimization. This integration allows us to define a feasible set of policies (π_C) that satisfy predefined safety constraints $J_{ci}(\pi) \leq d_i$, constrained Markov decision processes (CMDPs) incorporate a set of cost functions $c_1, \dots, c_k$, which are distinct from the reward function, as outlined in the framework of (CMDPs) [11]. In our scenario, if the agent makes contact with an obstacle at time step t , it incurs a cost of $c(s_t) = 1$; otherwise, the cost is 0. This is represented by $d_i = 1$ if the robotic arm violates the safety constraint, and $d_i = 0$ otherwise. This is essential to constrain hazardous behavior within the Safety Gym environment using a robotic arm agent. The Lagrangian technique introduces a composite loss function that integrates both reward-related and cost-related elements, to navigate the trade-offs between achieving task objectives and adhering to safety constraints:

$$\pi^* = \arg \max_{\pi \in \Pi_C} J_r(\pi) \quad (9)$$

where $\Pi_C = \{\pi : J_{ci}(\pi) \leq d_i, i = 1, \dots, k\}$

Here, $J_r(\pi)$ represents the reward-based objective, and each J_{ci} represents a cost-based constraint function, defined the same way as an expected return or average return metric, but utilizing c_i instead of the reward r . Additionally, each d_i serves as a threshold, which is a hyperparameter selected by the user. By integrating these methodologies, our research not only advances the field of policy optimization in RL but also significantly contributes to the development of safety-aware human–robot interactions. This ensures that robotic agents can operate within predefined safety margins in simulated environments. This dual approach, leveraging the strengths of PPO and its Lagrangian extension, forms the cornerstone of our implementation, promising to pave the way for innovative solutions in the realm of constrained RL, which is essential for learning safe policies in the real world.

4.1 Multi-step dense reward system

This section outlines the implementation of a multi-step dense reward system designed for training a robotic arm to perform object manipulation tasks. The reward system is structured to provide continuous feedback, gradually guid-

ing the agent through stages such as approaching the object, establishing contact, lifting, placing, and disposing of the object. The design of the reward structure draws on established principles in RL and robotics to ensure efficient task learning and execution.

Reward Function Formulation

The reward function r_t at time step t is defined as:

$$r_t = r_{\text{approach}} + r_{\text{contact}} + r_{\text{slippage}} \\ + r_{\text{lifting}} + r_{\text{placing}} + r_{\text{disposal}} + r_{\text{collision}}$$

where each component is defined as follows:

- **Approach Reward:**

$$r_{\text{approach}} = -\min(0.1 \cdot \|\text{tip} - \text{obj}\|, 5)$$

Encourages minimizing the distance between the end-effector tip and the object. This approach aligns with the principles of reward shaping to accelerate learning by providing continuous feedback [45].

- **Contact Reward:**

$$r_{\text{contact}} = \begin{cases} 5 & \text{if both fingers establish contact and no} \\ & \text{previous contact was established} \\ 1 & \text{if both fingers maintain contact} \\ 0 & \text{otherwise} \end{cases}$$

Rewards the agent for establishing and maintaining contact with the object, facilitating stable grasping behaviors.

- **Slippage Penalty:**

$$r_{\text{slippage}} = \begin{cases} -2 & \text{if slippage counter exceeds threshold} \\ 0 & \text{otherwise} \end{cases}$$

Penalizes the agent for slippage, encouraging stable grasping. This component is inspired by methods that penalize unstable grasps to promote robustness in manipulation tasks [46].

- **Lifting Reward:**

$$r_{\text{lifting}} = \begin{cases} 10 & \text{if object height exceeds threshold and} \\ & \text{phase is 'approaching'} \\ 0 & \text{otherwise} \end{cases}$$

Rewards successful lifting of the object, marking the transition to the placing phase.

- **Placing Reward:**

$$r_{\text{placing}} = -0.3 \cdot \|\text{obj} - \text{target}\|$$

Penalizes the distance to the target placement position, guiding the agent toward accurate placement. This dense reward formulation aids in reducing sample complexity during training [47].

- **Disposal Reward:**

$$r_{\text{disposal}} = \begin{cases} 20 & \text{if object is disposed successfully} \\ 30 & \text{if all objects are disposed successfully} \\ 0 & \text{otherwise} \end{cases}$$

Rewards successful disposal of the object and all objects, reinforcing task completion.

- **Collision Penalty:**

$$r_{\text{collision}} = \begin{cases} 1 & \text{if collision detected} \\ 0 & \text{otherwise} \end{cases}$$

Penalizes collisions to encourage safe interactions, aligning with safety-oriented reward shaping frameworks [48].

The reward structure balances multiple objectives: (1) task progression through phase-specific incentives, (2) precision requirements through variable penalty coefficients, (3) safety compliance through collision penalties, and (4) exploration efficiency through milestone-based rewards. This hierarchical approach enables the agent to learn complex manipulation behaviors while maintaining safety constraints throughout the learning process.

4.2 Neural network configuration for decision-making

Our research employs a TensorFlow-based multilayer perceptron (MLP) actor-critic model with two hidden layers of 64 neurons each to handle sequential decision-making in continuous action spaces. We implement two RL algorithms: PPO [49] and its Lagrangian variant (CPPO). PPO optimizes cumulative rewards using a clipped surrogate objective, while CPPO incorporates Lagrangian multipliers to enforce safety constraints during training. Both algorithms utilize on-policy learning with controlled policy rollouts (collecting data) and low-dimensional state information enhanced by tactile feedback through Grasp Mechanics (Fig. 2).

4.3 Model structure and training

- **Neural Network Architecture:** eight-dimensional input, two hidden layers with 64 neurons, four-dimensional output for actions.
- **Training Dynamics:** The agent was trained for 250 epochs, each comprising 2000 steps. Episode lengths

were capped at 500 and 1500 time steps for the respective tasks.

- **Policy Network:** MLP Gaussian policy for continuous actions calculates actions, log probabilities, and optimization metrics.
- **Value Networks:** Estimate state and state-conditioned values for accurate policy evaluation.
- **Hyperparameter Configuration:** Our implementation extends the Safety Gym framework [11] from mobile agent navigation to robotic arm manipulation with stringent safety control. Key modifications from Safety Gym defaults include reduced value function learning rate ($\text{vf_lr}: 1 \times 10^{-5}$), optimized steps per epoch (1000), and episode length (500) for manipulation tasks. Other parameters maintain Safety Gym's validated safe RL configuration:
 - $\text{penalty_lr}: 5 \times 10^{-2}$
 - $\text{gamma}: 0.99$, $\text{cost_gamma}: 0.99$
 - $\text{lam}: 0.97$, $\text{cost_lam}: 0.97$
 - $\text{ent_reg}: 0.0$, $\text{cost_lim}: 25$
 - $\text{penalty_init}: 1.0$, $\text{target_kl}: 0.01$
 - $\text{vf_iters}: 80$
- **Optimization Techniques:** Includes 80 gradient descent iterations per update cycle, employing a clipping ratio to mitigate drastic policy changes, and backtracking steps for efficient optimization.

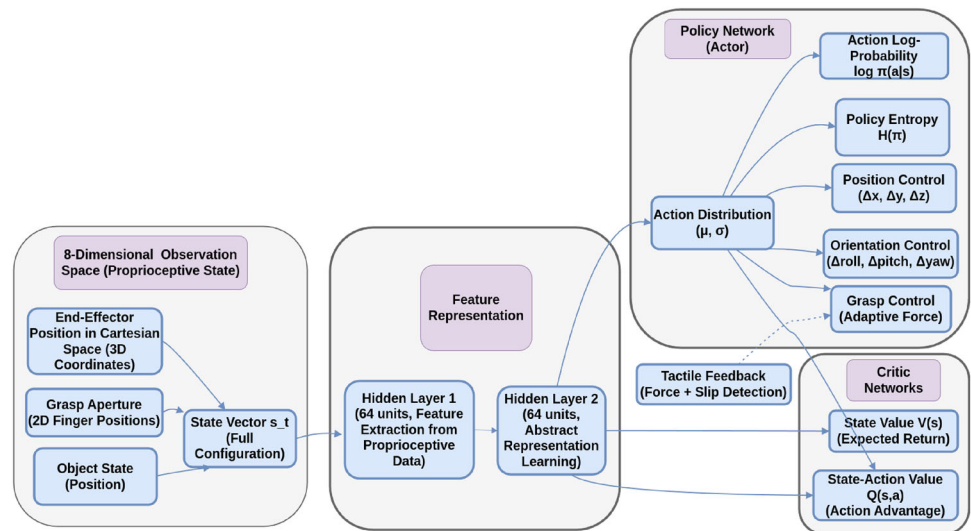
4.4 Grasp mechanics

Humans excel at manipulating unknown objects through tactile feedback, learning object properties during interaction without requiring detailed geometric knowledge [50]. This adaptive capability contrasts with traditional vision-based robotic systems that face limitations from occlusion, lighting variations, and processing latency in real-world scenarios.

Our research introduces a tactile-driven grasp mechanics framework that emulates human-like adaptive interaction through Reinforcement learning [51]. Rather than relying on preprogrammed object databases, our system learns object properties dynamically through tactile exploration using PyBullet's `getVisualShapeData` function and force sensing capabilities, enabling the agent to perceive shape type and dimensions while visual properties can be incorporated when needed. This approach treats each interaction as an opportunity for learning, developing understanding of surface friction, weight distribution, compliance, and geometric constraints through continuous tactile feedback.

Our tactile-driven framework integrates five core processes that enable sophisticated object manipulation through sensory-guided learning:

Fig. 2 Neural network architecture of the policy network



Algorithm 1: Constrained PPO

Data: Environment (env)

Result: Trained Policy

Initialization:

- Initialize logger, seeds, environment, placeholders, buffers, parameters, optimizers, and session

Policy Learning:

- Compute likelihood ratio, surrogate advantage, and cost

Main Loop:

- Initialize environment variables

Epoch Loop: for each epoch do

 Update penalty;

 for each step do

 Render environment;

 Get action from policy, step in environment;

 Store experience, update stats;

 if terminal state or end of epoch then

 Finish path, reset environment;

 end

 end

 Save model;

 Update policy and value functions;

 Log performance and stats;

end

Main Function:

- Parse arguments, initialize logger and agent
- Execute run_polopt_agent(parameters)

- **Force-Guided Orientation Optimization:** Determines optimal gripper alignment through contact force distribution analysis, maximizing grasp stability while accommodating object-specific constraints.
- **Contact-Based Position Targeting:** Establishes optimal gripper positioning through tactile exploration, ensuring precise alignment with secure grasp points discovered through sensory feedback.
- **Tactile Grip Validation:** Assesses grasp security through contact force analysis and slip detection before manipulation, preventing failures by ensuring adequate contact quality.

This tactile-centric approach enables human-like adaptive manipulation by prioritizing sensory feedback over preprogrammed knowledge. Combined with Safety Gym capabilities, our framework demonstrates that sophisticated grasping behaviors emerge through tactile learning, enabling robotic systems to navigate safety-critical manipulation tasks with enhanced precision and reliability.

4.5 Task success and evaluation metrics

Task success is evaluated using three metrics from the Safety Gym protocol [11]: (1) **Task Performance**, measured as the percentage of successfully completed tasks (approach, grasps, lift, place, and retract/done) out of total attempts (Table 1 and Figures 6,7), $\text{Success Rate} = \frac{\text{Successful Tasks}}{\text{Total Tasks}}$; (2) **Constraint Satisfaction**, tracking safety violations (e.g., collisions), with fewer violations indicating better safety; and (3) **Average Regret**, reflecting cumulative safety costs during training, $\text{Average Regret} = \frac{1}{N} \sum_{i=1}^N C_i$, where C_i is the safety cost at step i , and N is the total number of steps. These metrics balance task efficiency and safety compliance.

Table 1 Performance summary across evaluation objects

Object	Success Rate (%)	Completion Time (s)	SOPM Success 1-Obj Episode	MOAM Success 3-Obj Episode
Lipton Tea Box*	99.0	13.0	39/40	
Cube*	99.0	13.32	39/40	
Tea Cup	95.0	13.67	36/40	32/40
Soap Bar	95.0	13.10	36/40	
Remote Control	85.0	13.74	35/40	
Cleanser Bottle	60.0	14.0	25/40	

Note: Success Rate and Completion Time are averaged across both SOPM and MOAM tasks. SOPM involves single-object manipulation per episode, while MOAM involves multi-object manipulation per episode. The success counts reflect the number of successful objects (SOPM) and successful episode manipulations (MOAM) out of the total attempts

4.6 Zero-shot policy transfer across objects

In the SOPM and MOAM tasks, our study focused on training a robotic system to manipulate a Lipton tea box and a cube, respectively, through an extensive 500,000-step learning process, with the goal of evaluating the system's zero-shot generalization capabilities to unseen objects. Using a multilayer perceptron actor-critic model (MLP), policies were developed based on essential low-dimensional state information.

To assess generalization, we tested the trained policies on four novel objects: a teacup, a soap bar, a remote control, and a cleanser bottle. The system successfully grasped all of these objects, effectively adapting to variations in shape and orientation. However, the cleanser bottle presented unique challenges due to its irregular geometry, particularly the offset of its dispensing cap, which introduced alignment issues for the gripper and resulted in a lower success rate.

These results suggest that, while the system demonstrates strong generalization to standard object geometries, additional refinement may be needed to handle more complex or asymmetrical shapes. Throughout the experiments, the system consistently adhered to safety constraints, demonstrating robust policy performance and highlighting its potential to generalize safely across a wide range of objects without explicit prior training.

4.6.1 Systematic evaluation

Figure 3 shows the robotic arm successfully manipulating unseen objects: a teacup, a soap bar, remote control, and a cleanser. The grasping and maneuvering actions for the teacup, soap bar, and remote control were performed in the preferred way without any additional training. However, the cleanser presented a challenge due to its offset at the edge of the tap. All actions adhered to safety constraints.

4.6.2 Performance metrics

The policy achieved a success rate of 99% on the training objects, demonstrating strong performance in familiar scenarios. When evaluated on unseen objects, including a teacup, soap bar, remote control, and cleanser, it maintained a success rate of 84% for the teacup, soap bar, and remote control, highlighting its ability to generalize to objects with relatively simple geometries. However, the success rate dropped to 60% for the cleanser, mainly due to the misalignment of the gripper caused by the offset edge of the tap in the design of the cleanser. This decline suggests that current policy struggles with more complex or irregular shapes, particularly when precise alignment is required within constrained time frames.

Overall, these results indicate that while the policy generalizes well to standard objects, further refinement is necessary to improve robustness against non-uniform geometries, especially in safety-critical tasks, while continuing to adhere to strict safety constraints throughout the manipulation process (Fig. 4).

4.6.3 Comparative analysis

We present a comparative analysis of our closed-loop grasping approach against traditional fixed-policy methods [14, 34] and state-of-the-art RL frameworks such as Panda-Gym [18]. While Panda-Gym enables goal-conditioned learning on the Franka Emika Panda robot, it primarily targets short-horizon, simplified manipulation tasks and does not incorporate explicit safety constraints—an essential factor in real-world human–robot collaboration settings.

In contrast to fixed-sequence methodologies [14, 34], our method adopts a fully learning-based framework that enables dynamic adaptation through a sequential decision-making architecture. A key distinction lies in the system's ability to monitor environmental feedback and replan accordingly. For instance, when a grasp attempt fails, the agent reevaluates

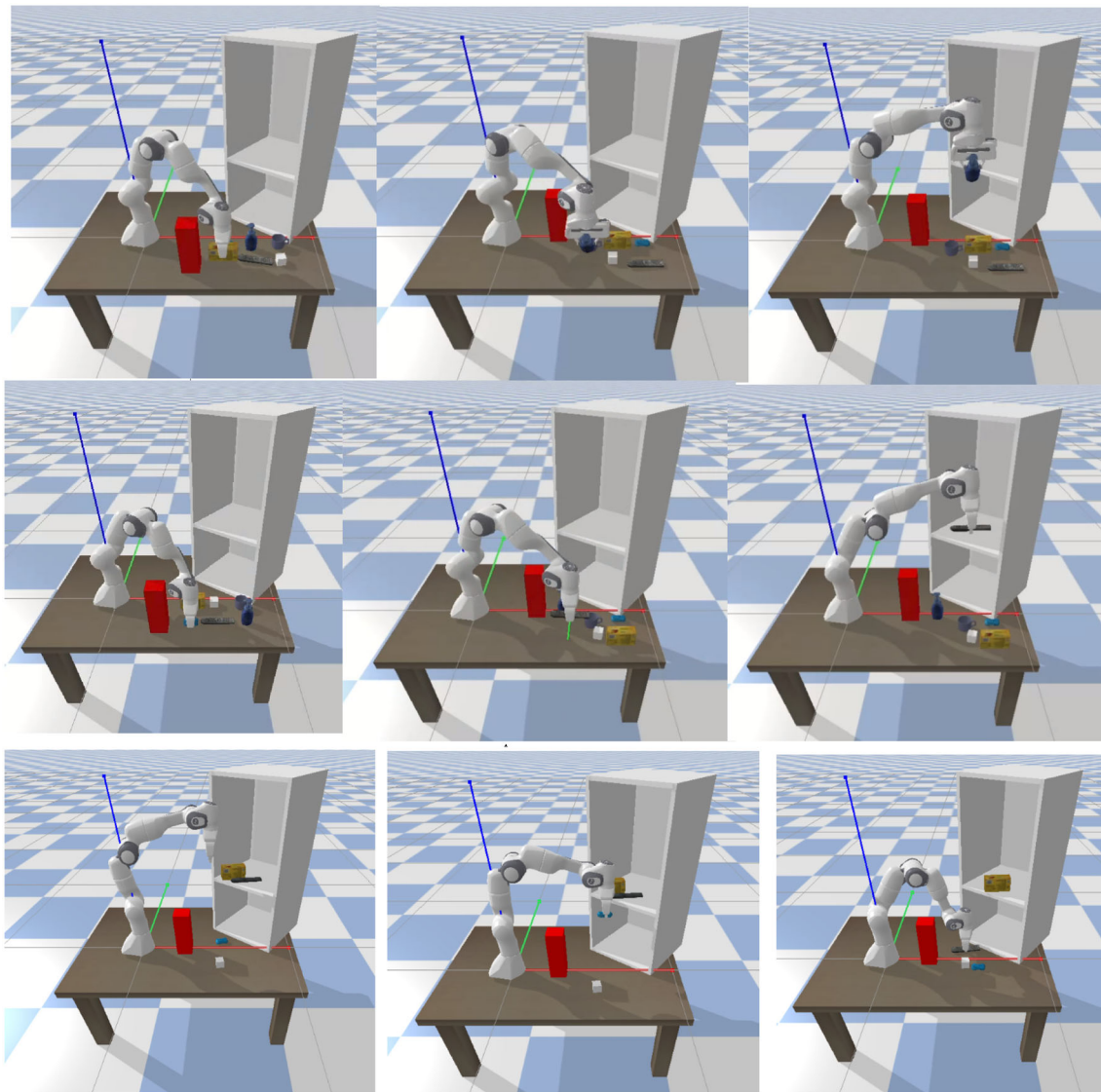


Fig. 3 Zero-shot policy transfer across objects while the robot maneuvers around obstacle

and reattempts grasping, instead of executing a rigid, pre-programmed sequence. As illustrated in Fig. 3 and detailed in Table 2, our method achieves higher success rates and demonstrates enhanced adaptability to varying object shapes and positions.

Comparison with Existing Approaches We empirically benchmarked our method against existing baselines across several key features, as summarized in Table 2.

4.6.4 Supplemental evaluation

Safety Gym on Handoversim Benchmark:

In our supplementary study, we evaluated the integration of the Safety Gym framework with the Handoversim benchmark to enhance safety in human-to-robot object handover tasks. The trained policy includes planning, approaching, and

attempting to grasp objects. In our current Handoversim simulation, the robotic agent operates autonomously, driven by a learned policy with adaptive action selection based on real-time state inputs and environmental feedback. In contrast, the human hand's motion is predefined using high-fidelity trajectories from human demonstrations, incorporating DexYCB object models [13] and the deformable MANO mesh model [52]. The hand and object follow a synchronized 6D trajectory, removing the need for a fully accurate hand model while preserving natural movement timing.

We acknowledge that the current setup lacks bidirectional, real-time adaptive interaction, a key feature of realistic human–robot collaboration (HRC). The human hand movement is non-interactive, derived from predefined data and not responsive to the robot's actions. In contrast, the robot autonomously adapts to the incoming object and adjusts its

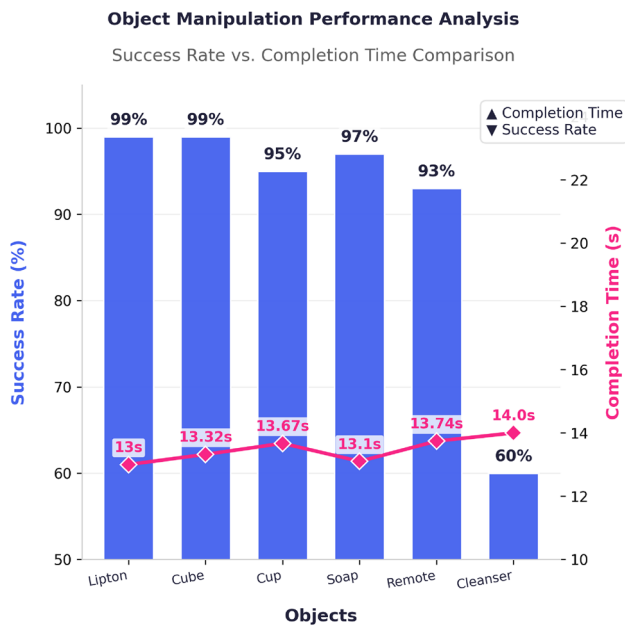


Fig. 4 Zero-shot policy transfer performance across diverse objects. Results represent the average success rate over 40 trials per object

grasp based on the object's position and configuration. Thus, our setup is best described as a unidirectional handover simulation, where the robot demonstrates adaptive behavior in response to a fixed human hand trajectory.

Despite significant advancements in the Handoversim benchmark, fully autonomous end-to-end policy learning remains challenging and often necessitates switching to hand-coded policies for effective grasping. Our findings suggest a promising shift in paradigm, such as the implementation of a teacher–student training framework. This framework is crucial for direct learning in end-to-end RL policies, particularly in safety-critical applications where real-world risks must be mitigated.

This supplemental evaluation highlights how the Safety Gym framework facilitates safe and efficient human-to-robot

interactions, underscoring the importance of safety in practical applications. It also supports our main research by demonstrating the potential of safe RL to overcome current limitations in policy learning for robotic handovers (Fig. 5).

5 Experiment design and realization

In our analysis of robotic arm control within a 3D Cartesian space, we found that increasing the dimensionality of the state vector does not significantly improve the agent's learning efficiency for adaptive grasping. However, tactile feedback does improve the agent's learning efficiency and effectiveness, as previously discussed in the 'Grasp Mechanics' section.

In a comparative study of proximal policy optimization (PPO) and its constrained version, cPPO, across two manipulation tasks of increasing complexity, we observed that cPPO, which integrates Lagrangian safety methods, results in lower average costs, suggesting that cPPO achieves safer performance, although it requires a slightly longer training period compared to PPO.

Figure 6 illustrates the performance of these algorithms, emphasizing the safety-efficiency trade-off. The graph tracks reward values over training cycles, with the horizontal axis representing the training cycles and the vertical axis showing learning cycle-wise reward values. It compares the average cumulative rewards and average costs per episode for both PPO and cPPO, effectively demonstrating the safety and efficiency considerations when using constrained optimization in robotic control.

This visual comparison highlights that cPPO, while exhibiting a more gradual learning curve, consistently maintains lower cost metrics. This behavior suggests that cPPO prioritizes safety, as reflected by its systematically minimized cost values, thereby ensuring safer operational dynamics for the robotic arm.

Table 2 Comparison of object grasping and manipulation approaches

Feature	Our Approach	Baselines [14, 34]
Learning Paradigm	Goal-directed RL	Imitation learning with predefined behaviors
Decision Making	Sequential, state-dependent with learned policy	Fixed action sequences with predefined policy
Failure Handling	Dynamic reassessment with real-time adaptation	Continuation with predefined sequence, static behavior
State Awareness	Continuous monitoring of gripper state	Limited state feedback
Task Execution	Safety-aware operation while maintaining performance	Performance optimization without explicit safety considerations

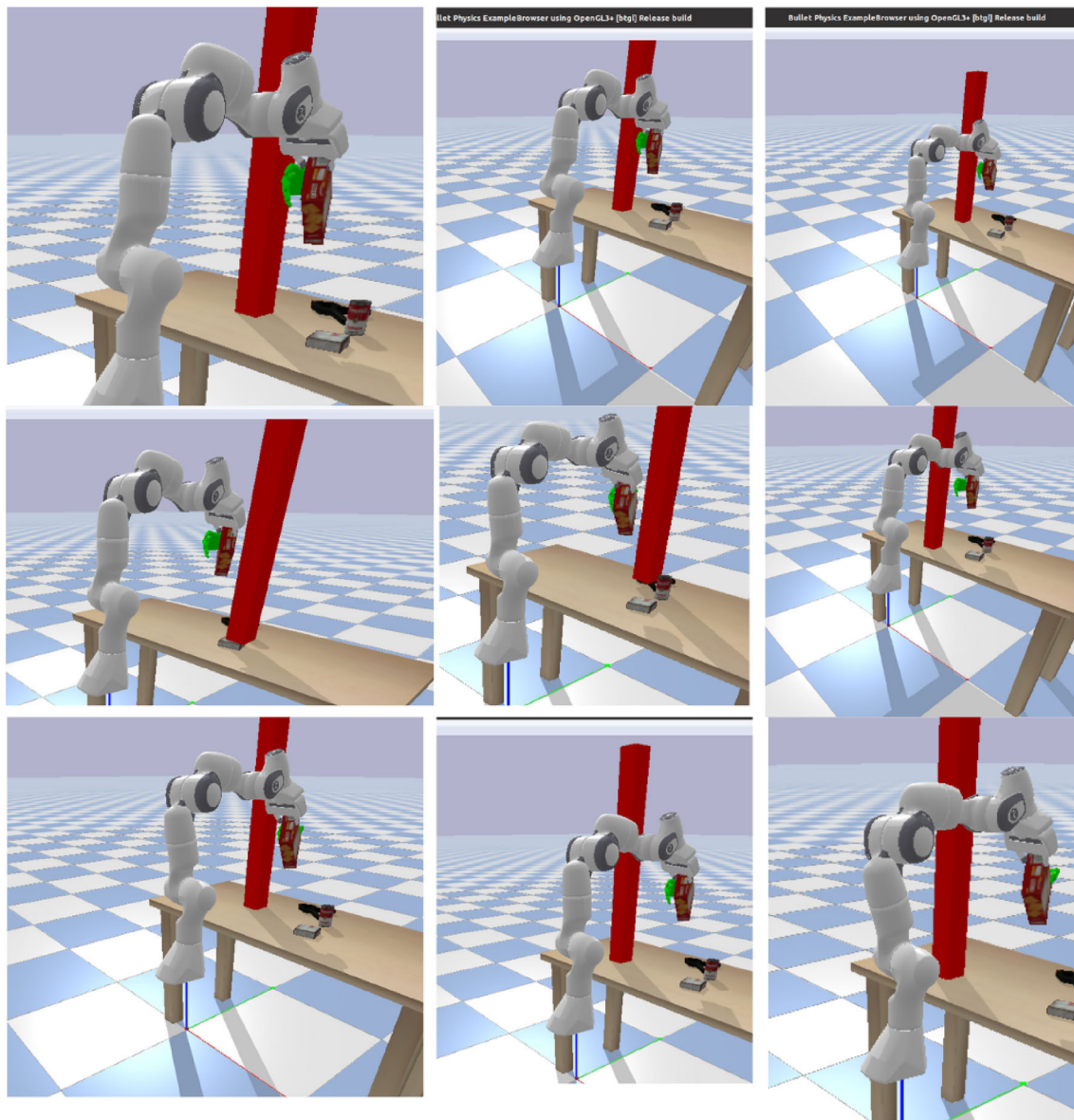


Fig. 5 This figure depicts the sequential actions of a robotic arm performing a grasping task in the Handoversim environment. The robot maneuvers around an obstacle while attempting to grasp an object, showcasing the simulation's handling of safety-critical task operations

In the MOAM environment, Figure 7, both PPO and cPPO algorithms demonstrate steady improvement in performance, achieving a maximum success rate of approximately 7.5. This is notably lower than the success rates observed in SOPM, indicating a slower learning progression. The gradual ascent in the learning curve reflects the increased complexity inherent in manipulating three objects consecutively within a single episode, without resetting. The extended episode length of 1500 time steps provides the agent with more opportunities to refine its policy; however, the added complexity results in a more gradual learning trajectory compared to SOPM.

Regarding constraint violations, cPPO consistently outperforms PPO across both tasks, maintaining a lower average episode cost throughout training. In the MOAM environment, while both algorithms initially exhibit higher violation costs, cPPO promptly reduces these violations, demonstrating its effectiveness in constraint management while exhibiting a comparable, yet safer, policy. This trend underscores the challenges associated with managing multiple objects simultaneously in each episode. The higher constraint violations in MOAM reflect the increased difficulty in handling the intricate interactions among multiple objects, whereas SOPM benefits from simpler task interactions, allowing for better constraint handling.

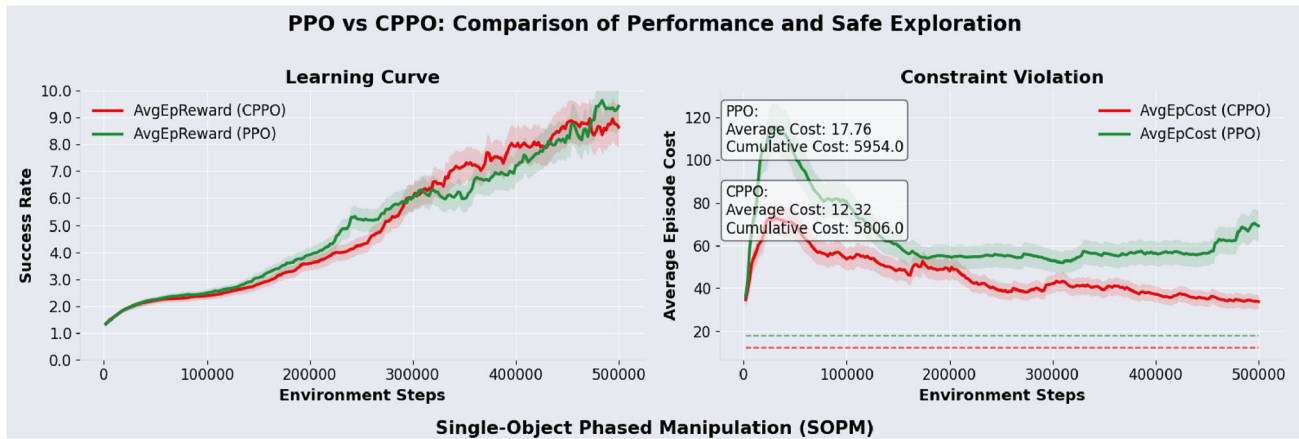


Fig. 6 PPO vs. cPPO Rewards and Costs Comparison in Object Manipulation

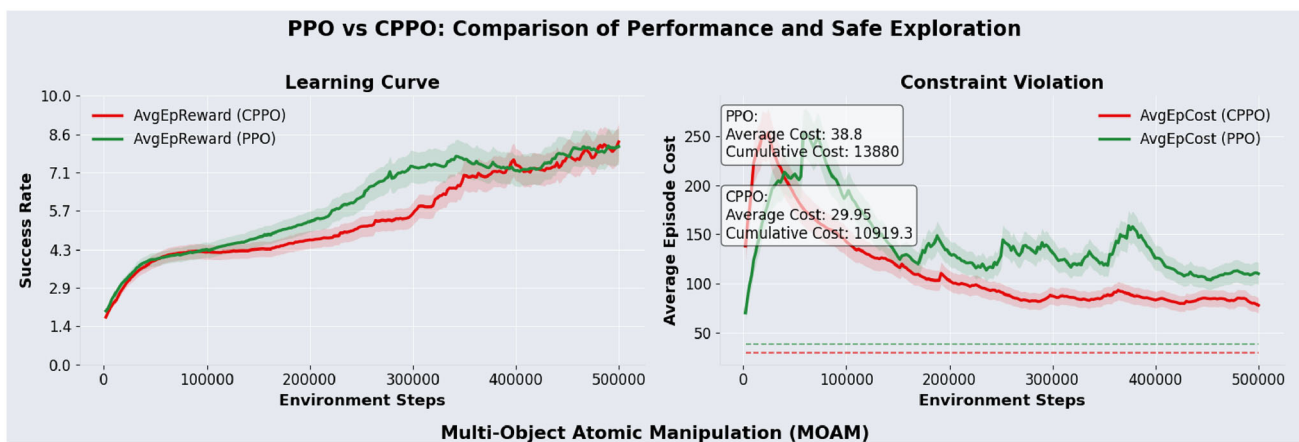


Fig. 7 PPO vs. cPPO Rewards and Costs Comparison in Object Manipulation

Table 3 Evaluation of PPO and cPPO performance in terms of reward and safety metrics across two distinct manipulation tasks

Task	Algorithm	Final Avg Reward %	Final Avg Cost	Cumulative Cost
SOPM	PPO	9.5	17.76	5954.0
	cPPO	8.8	12.32	5806.0
MOAM	PPO	7.5	38.80	13880
	cPPO	7.2	29.95	10919

Table 3 summarizes the final reward and safety metrics for PPO and cPPO across the two tasks. While the cumulative cost differences are relatively small, this metric reflects all agent behavior, including early unsafe exploration. In contrast, the average episode cost captures recent performance and safety behavior more accurately. Since cPPO learns to reduce violations over time, its average cost decreases significantly, showing improved long-term safety compliance, even if cumulative cost still reflects early errors

- **Context:** The table shows the average comparison costs per run for the standard PPO algorithm and its constrained version (cPPO).
- **Task:** The task involves manipulating various geometries and securing them at the target location, all while adhering to stringent safety constraints.
- **Method:** 3D action representation in Cartesian space.

5.1 Supplemental evaluation

Safety Gym on Handoversim Benchmark: Similar to our custom simulation results presented above, we compare the evaluation metrics for PPO and cPPO using the Handoversim benchmark. In Handoversim, we achieved similar performance only in attempting to grasp objects. However, the agent rarely succeeds in grasping the object, as it relies on an end-to-end learning policy. Established baselines in

Handoversim switch to hand-coded policies for stable and reliable grasps, highlighting the limitations of end-to-end policy learning in this context. Unlike our custom simulation, Handoversim does not include extended sub-tasks due to previously mentioned inherent simulation challenges.

This visual comparison shows, the PPO algorithm's performance over time is a steady increase in average episode reward, reflecting the algorithm's learning and improvement. The cPPO algorithm starts with a slightly higher reward and maintains better performance compared to PPO, demonstrating more significant improvement over time.

Regarding cost, initially, the PPO costs are positive but gradually decrease as the algorithm learns. The dashed red line represents the mean cost over time, indicating an average cost of -0.13 per episode, with a cumulative cost of -26.85 . Conversely, the cPPO algorithm starts with a higher cost but reduces more significantly over time. The dashed purple line represents the mean cost, indicating an average cost of -0.21 per episode, with a cumulative cost of -41.4 (Fig. 8).

Performance: Both algorithms improve their performance over time, as indicated by the increasing average episode rewards. The cPPO algorithm demonstrates a more substantial improvement and consistently performs better than PPO, achieving higher rewards.

Cost Management: The cPPO algorithm manages costs better than PPO, as indicated by its lower (more negative) average and cumulative costs. This suggests that cPPO not only learns to achieve higher rewards but also does so more efficiently by minimizing costs.

5.2 Discussion and conclusion

This paper addresses a critical yet often overlooked challenge in robotic manipulation: ensuring safety in realistic object manipulation scenarios. We present a novel approach that integrates compact proprioceptive state acquisition with fine-tuned control and tactile feedback, enhanced by Grasp Mechanics within the Safety Gym framework. Through DRL, our method tackles the fundamental challenges of safe and precise manipulation across objects with diverse geometries while relying exclusively on low-dimensional kinematic data, eliminating the dependency on detailed visual data, providing a significant advantage in real-world environments where such data may be unavailable or unreliable. A key contribution of our research lies in its emphasis on learning-based goal-directed tasks and the system's demonstrated adaptability to unfamiliar objects, a crucial capability for deployment in dynamic and unpredictable environments. Despite employing a simple two-finger gripper configuration, our system exhibits efficient navigation and reliable grasping techniques across a diverse range of objects, rendering our approach particularly suitable for safety-critical

applications where precise navigation around obstacles and secure manipulation are paramount.

Our comprehensive performance analysis reveals that constrained proximal policy optimization (cPPO) effectively balances exploration with safety constraints, demonstrating significant value in environments requiring careful obstacle navigation. The algorithm's gradual learning progression and inherent safety mechanisms position cPPO as a strong candidate for real-world applications where operational risks must be minimized. In SOPM task, both PPO and cPPO demonstrate steady improvement trajectories, with PPO achieving success rates exceeding 90% and cPPO reaching 90%. The rapid learning convergence reflects the relative simplicity of single-object manipulation, facilitated by shorter training episodes (500 time steps) that enable faster feedback loops and accelerated convergence. Conversely, in MOAM task, both algorithms achieve lower success rates of approximately 78%, indicating the inherent complexity of multi-object scenarios. This performance degradation stems from the increased task complexity involving three objects within extended episodes (1500 time steps), resulting in more intricate environmental interactions and prolonged learning phases.

Regarding constraint violations, cPPO in SOPM consistently outperforms PPO with fewer safety violations, while in MOAM, the performance differential narrows considerably as cPPO demonstrates improvement over extended training periods. This convergence reflects the heightened challenges associated with simultaneous multi-object management and highlights a fundamental trade-off between task complexity and learning efficiency: SOPM facilitates faster convergence and higher success rates due to its simplified structure, while MOAM presents challenges but enables more sophisticated manipulation capabilities. While our algorithm demonstrates effectiveness in simulated environments, the transition to real-world implementation presents significant challenges inherent to the simulation-to-reality gap, a persistent issue in RL applications. To address these challenges, our future research will leverage the teacher–student training framework, an established methodology where policies are initially learned using simplified, low-dimensional state representations before transfer to high-dimensional inputs such as raw imagery [19]. This hierarchical approach involves training a “teacher” model on low-dimensional kinematic data to establish baseline task understanding, subsequently guiding a “student” model trained on complex visual inputs, significantly reducing computational overhead while improving sample efficiency compared to direct vision-based learning approaches.

Our approach presents two primary limitations that warrant consideration in the context of broader deployment scenarios. First, the implementation of a fixed orientation strategy, rather than full 6-DOF control, constrains the

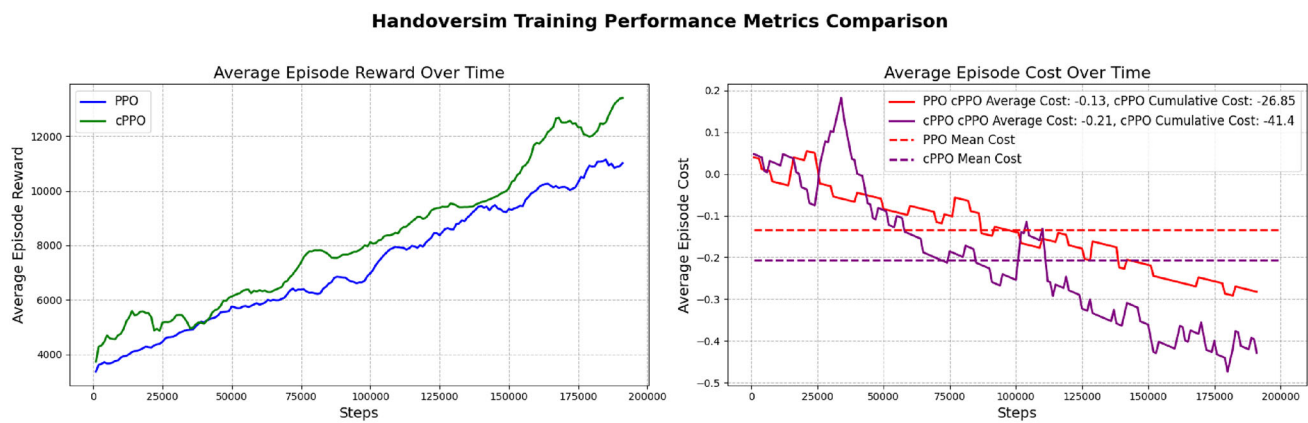


Fig. 8 This graph provides a comprehensive comparison of the performance and cost management capabilities of the PPO and cPPO algorithms in a Handoversim training environment, indicating that cPPO tends to have better performance in terms of both reward and cost efficiency

system's ability to execute orientation-sensitive grasps, particularly for objects with asymmetric affordances (e.g., a shampoo bottle with the pump facing upward), which cannot be effectively grasped from a top-down view. While this design choice reduces learning complexity and maintains practical effectiveness across common household objects, it limits manipulation versatility for specialized tasks requiring precise orientation control. Second, exclusive reliance on proprioceptive feedback, while computationally efficient and theoretically more transferable to real-world scenarios, may limit performance in tasks requiring visual guidance for precision manipulation. However, our results demonstrate that proprioceptive sensing enables robust grasping while maintaining strict safety adherence, providing a solid foundation for future extensions. Two promising avenues emerge for future investigation: expanding to full 6-DOF control through hierarchical policy architectures that initially master 3D positional control before incorporating dynamic orientation adjustments, and implementing the teacher–student training framework where our safe manipulation policies serve as instructors for vision-based student networks, potentially reducing computational demands while maintaining safety guarantees with significant implications for real-world deployment.

Our research advances autonomous robotics by demonstrating that effective object manipulation can be achieved while maintaining strict safety compliance through the integration of constrained RL with proprioceptive feedback, establishing a promising foundation for simulation-to-real transfer. Unlike vision-dependent approaches that often suffer from the reality gap due to visual domain discrepancies, our proprioceptive-focused methodology relies on physical interactions and low-dimensional representations that exhibit superior theoretical transferability across domains. This design philosophy directly addresses a critical challenge in robotics: the significant performance degradation

experienced by simulation-trained policies when deployed on physical hardware. By prioritizing contact dynamics, force-based feedback, and physics-grounded policy constraints, our approach demonstrates inherent resilience to domain transfer challenges that typically impede real-world deployment success. Future work will systematically address the simulation-reality gap through domain randomization and progressive transfer techniques [33, 53], with particular emphasis on preserving safety guarantees while enhancing sample efficiency across the complete spectrum of manipulation tasks—from controlled laboratory environments to dynamic human-centered settings.

Ultimately, this work represents a significant advancement toward enabling robots to interact safely and effectively in human-centered environments. By emphasizing adaptability, safe obstacle navigation, and computational efficiency, our research establishes a foundational contribution to autonomous, safety-critical human–robot interaction, paving the way for more robust and reliable robotic systems in real-world applications. The demonstrated capability to maintain safety constraints while achieving effective manipulation performance positions this approach as a viable pathway for deploying robotic systems in environments where human safety and operational reliability are paramount concerns.

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Author Contributions All authors contributed significantly to the conception and design of the study. Wei Feng, Huang Tianlun, and Wang Zhiyong organized weekly meetings, during which they engaged in constructive discussions, critically assessing progress and exploring ways to

enhance the study's approach. Shahid Asad Ali provided valuable guidance on conducting literature research, identifying gaps in the existing knowledge, and ensuring a comprehensive review. Liu Xiao, as a senior researcher, offered expert advice on the implementation of the study and assisted in finding technical solutions. Wang Weijun contributed to the supervision of the study, reviewed the final draft, highlighted overlooked aspects, and managed funding.

Data Availability There are no shareable data associated with this paper.

Code Availability The primary codebase used in this study is publicly available on GitHub (<https://github.com/openai/safety-starter-agents>) and has also been discussed in the reference section [11]. Specific implementation and modifications used in this study are not publicly available but can be provided upon request.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

Consent for publication All authors have given their consent for the publication of this manuscript.

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